



Online Retail Business

Customer Segmentation: From Raw Transactions to a Revenue-Weighted Growth Plan

A behavioral segmentation model built on 909 active customers, validated end-to-end — from data cleaning through to prioritized, revenue-backed recommendations.



909

Customers Segmented



3

Behavioral Segments



0.432

Silhouette Score



\$528.5K

Est. Total Monetary Value

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Prepared for internal team review

What We'll Walk Through

01

The Business Question

Why segment, and what decision this should drive

02

How the Model Was Built

Data cleaning problems found, feature engineering, model selection

03

The Three Segments

Who they are, statistically and behaviorally

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The Hidden Truth

Why customer count and revenue tell different stories

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Problems, Solutions & Business Plan

Root causes, fixes, owners, and a quarterly target timeline

06

Roadmap, Forecast & Impact

Prioritized actions, growth/decline scenarios, and projected upside

Not All Customers Deserve the Same Strategy

Treating every customer with one blanket marketing and retention strategy wastes budget on customers who will never convert, and under-serves the customers who already drive most of the revenue. The goal of this project was to answer three questions with data, not intuition:

1

Who are our customers, behaviorally?

Group customers by how they actually buy — not assumptions — using recency, frequency, spend, and engagement signals.

2

Where does our revenue really come from?

Weight each group by size AND spend — a large group isn't automatically the most valuable one.

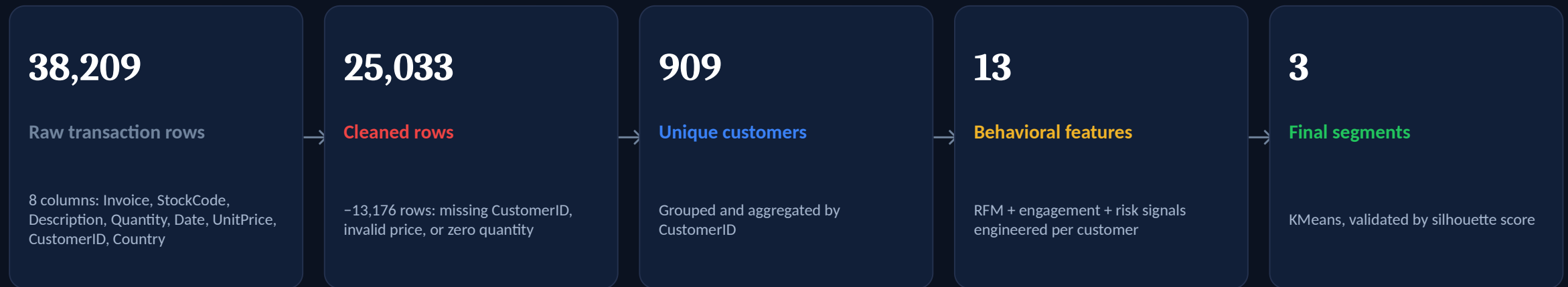
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What should we do differently for each group?

Turn the segmentation into specific, owned actions — not just a report that sits in a drive.

From Raw Transaction Logs to Customer Profiles

The source data is a UK online-retail export — one row per line item, not per customer. Before any modeling, it had to be cleaned and re-shaped into one behavioral profile per customer.



Why nearly a third of rows were dropped: segmentation is done at the customer level, so line items without a CustomerID (guest checkouts / unlinked orders) can't be attributed to a profile and were excluded, along with invalid price/quantity rows. Worth flagging: roughly 1 in 3 raw transactions could not be tied to a known customer.

Data Quality — What We Found While Cleaning

Before trusting any downstream number, every raw column was checked. Four concrete problems surfaced — each handled differently depending on whether it blocks customer-level analysis.

Problem	Rows Affected	% of Raw Data	Action Taken
Missing CustomerID	13,174 rows	34.5%	Dropped — guest/unlinked orders can't be attributed to a customer profile
Missing Description	117 rows	0.3%	Kept — product-label gap only, doesn't block customer-level modeling
Missing / invalid UnitPrice	~1–2 rows	<0.1%	Dropped — can't compute TotalPrice without a valid price
Zero Quantity (returns/adjustments)	included above	—	Dropped — zero-quantity rows contribute no purchase signal
Missing Country	1 row	<0.1%	Kept — irrelevant to the 13 behavioral features used



Net result: 25,033 of 38,209 raw rows retained (65.5%), covering 909 unique, fully-attributable customers. The only material loss is the missing-CustomerID gap — worth a separate conversation with the checkout/CRM team about capturing customer identity more consistently at the point of sale.

13 Behavioral Features, Three Signal Families

Every customer's raw purchase history was compressed into a fixed set of behavioral signals, grouped into three families that together describe how, how often, and how safely a customer buys.



Value & Frequency

- Recency
- Frequency
- Monetary
- Total Quantity
- Average Order Value
- Average Quantity / Order
- Average Unit Price



Engagement & Loyalty

- Unique Products
- Customer Lifetime (days)
- Average Days Between Orders
- Purchase Rate
- Weekend Purchase Ratio



Risk

- Cancellation Rate

Choosing k — Why 3 Segments, Not 2

Silhouette score tested for k = 2 through 10 on the scaled 13-feature space.



k = 3 SELECTED

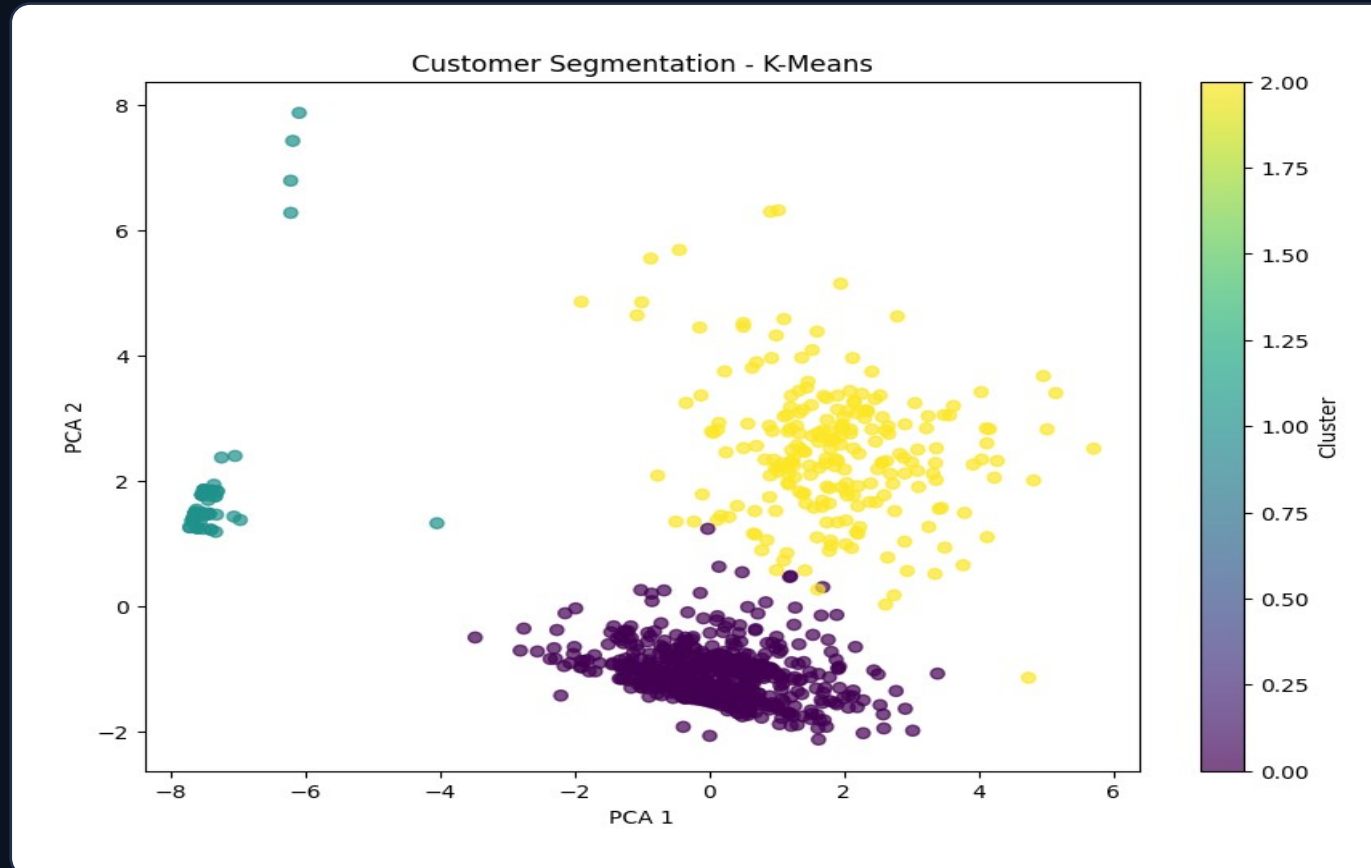
Silhouette 0.432 — best statistical separation among cluster counts that still produce an actionable, multi-group segmentation.

Why not k = 2, the higher score?

k = 2 scores higher (0.583) but collapses into one "everyone else" group — it can't separate healthy regular customers from the near-zero-value At Risk cohort. k = 3 trades a little statistical tightness for a segmentation that maps directly to three different business responses: protect, grow, investigate. k = 4–10 all score below 0.41.

Cluster Separation Holds Up Visually

Clustering itself used all 13 scaled features — PCA (2 components) is shown here purely to visualize the result in 2D. PC1 and PC2 together explain 67.4% of total variance (PC1: 43.3%, PC2: 24.1%), enough for this view to be a faithful picture of the underlying separation.



Three Segments, One Blended Customer Base



Segment	Customers	Avg. Monetary	Cancel. Rate
High Value	227 (25.0%)	\$1,272.93	3.51%
Regular	619 (68.1%)	\$386.98	0.50%
At Risk	63 (6.9%)	\$0.20	99.47%

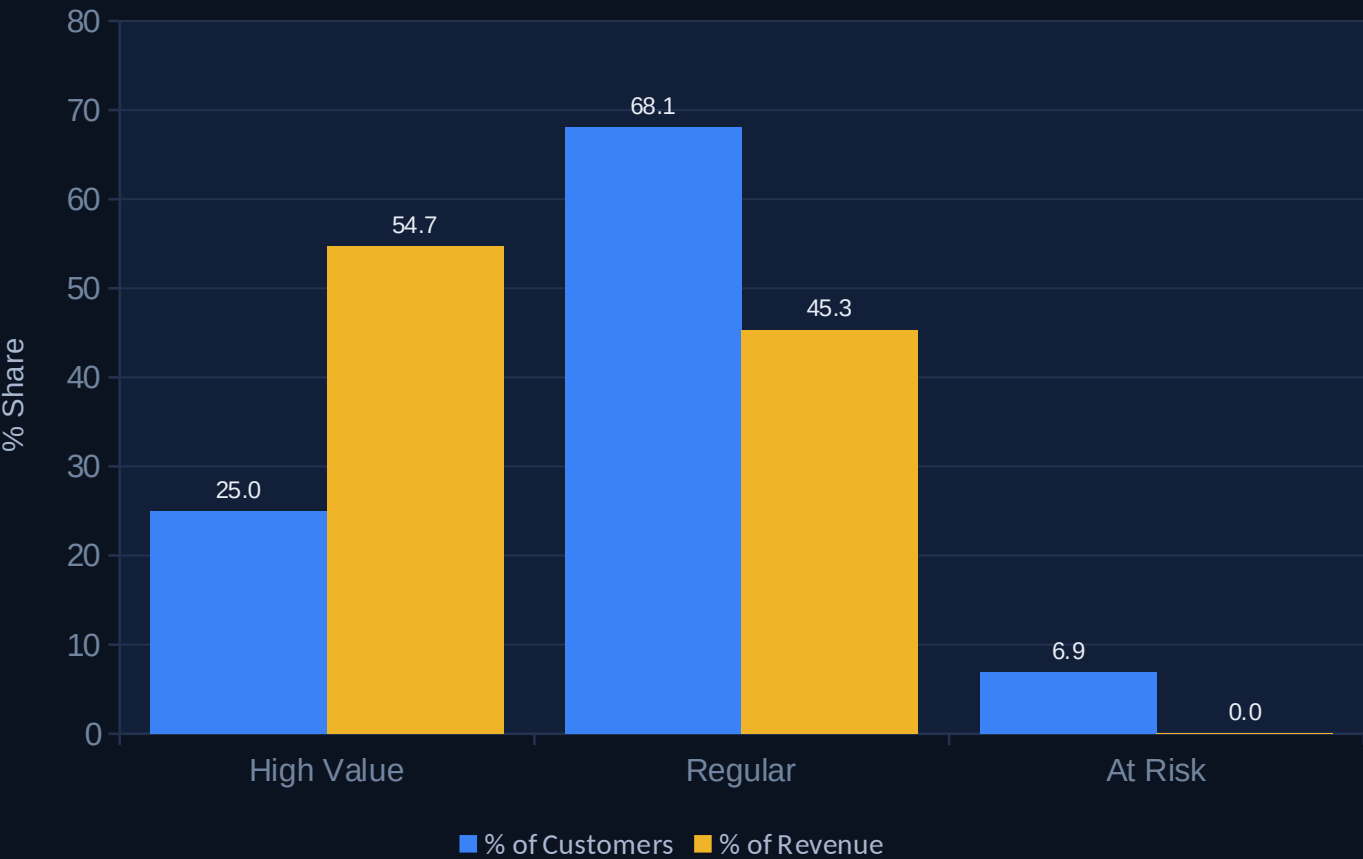
Regular — the majority by headcount; steady but low-frequency buyers.

High Value — a quarter of customers, buying 2.8x more often with 60% more product variety.

At Risk — small in headcount, but its 99.47% cancellation rate is a red flag, not typical churn.

Customer Count ≠ Revenue

Weighting each segment's average spend by its size tells a very different story than headcount alone.



99.47%

cancellation rate in the At Risk segment — 63 customers whose orders almost never convert into real revenue.

\$288,954.94

generated by High Value customers alone — 54.7% of estimated total revenue from just 25% of the base.

At Risk May Not Be a Marketing Problem At All

A 99.47% cancellation rate is not typical "at risk" churn behavior. Churn looks like declining frequency and rising recency — this looks like payment failures, bot/fraud activity, or a checkout bug affecting this cohort specifically.



Audit payment gateway logs

Check this segment's transactions specifically for repeated authorization failures.



Check for fraud signals

Shared IP addresses or device fingerprints across the 63 accounts would confirm bot/fraud activity.

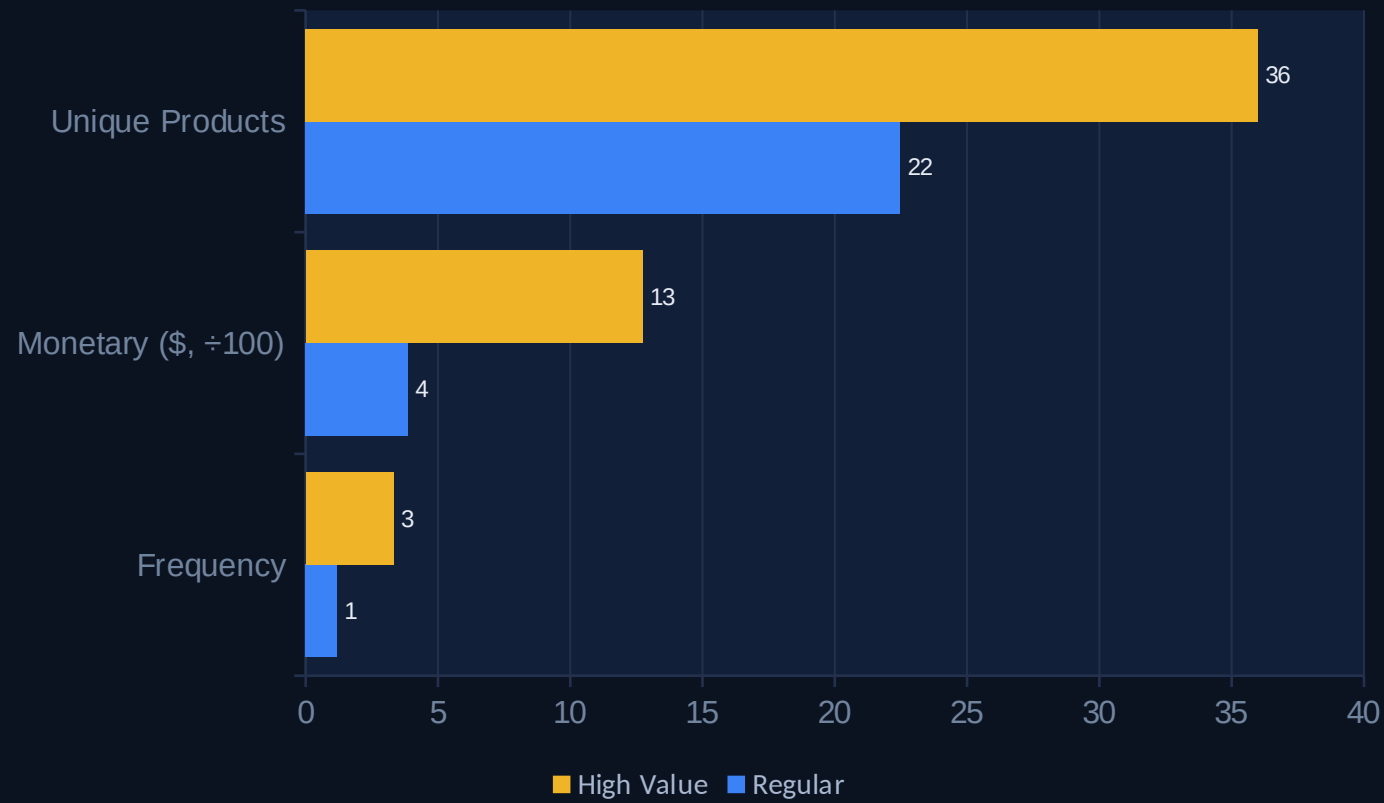


Exclude from retention campaigns

Don't spend win-back budget on this group until the root cause is identified — it won't convert.

Turning Regular Into High Value

High Value customers buy 2.8x more often, spend 3.3x more, and engage with 60% more unique products than Regular customers. Closing even a slice of that gap compounds fast across 619 people.



\$886

average monetary gap per customer (High Value – Regular)

+\$54,800

estimated revenue from closing just 10% of this gap across all 619 Regular customers — with zero new acquisition spend.

Problems, Root Causes & Solutions

Turning the findings above into a structured business-analyst view before we get to the roadmap.

Problem	Business Impact	Recommended Solution	Expected Outcome
At Risk: 99.47% cancellation rate (likely fraud / payment / checkout bug, not churn)	\$0.20 avg. value from 63 customers (6.9% of base) — inflates headcount health metrics while contributing ~\$0 real revenue	Audit payment gateway logs & device/IP fingerprints; exclude from retention spend until root cause is confirmed	Recover \$12K–\$25K/yr if fixable, or cleanly write off and stop wasting marketing budget on it
Regular segment under-converts vs. High Value (3.3x spend gap)	68.1% of customers generate only 45.3% of revenue — the largest pool of customers is also the most under-monetized	Targeted nurture track: bundles, loyalty points, 3rd/4th-order incentives for above-median-frequency Regulars	+\$55K–\$90K/yr from shifting 10–15% of Regular spend toward High Value behavior
34.5% of raw transactions have no CustomerID (guest/unlinked orders)	Real customer value is likely understated — these transactions are invisible to any customer-level model, this one included	Incentivize account creation / loyalty sign-up at checkout; tighten CustomerID capture in POS & e-commerce flow	More complete future models, more accurate segment sizing and revenue attribution
Segmentation is a single snapshot (will drift as behavior changes)	Campaigns risk targeting stale segments within 1–2 quarters as customers migrate between groups	Re-run clustering quarterly on the same 13 features; track segment migration as a health metric	Sustained targeting accuracy and early detection of decline before it shows up in revenue

From Insight to Action — Four Priorities



Investigate At Risk

Root-cause the 99.47% cancellation rate before treating it as a marketing segment.

Owner: Payments / Fraud



Protect High Value

227 customers drive 54.7% of revenue on a 3.5% cancellation rate — the healthiest segment by far.

Owner: CRM / VIP Program



Nurture Regular to High Value

Target above-median-frequency Regular customers with bundles, loyalty points, and 3rd/4th-order incentives.

Owner: Lifecycle Marketing



Re-run Segmentation Quarterly

Behavior shifts seasonally — refresh the model and track segment migration as a health indicator.

Owner: Data / Analytics

Turning the Roadmap Into a Quarterly Plan

Each priority gets an owner, a concrete target, and a quarter to hit it by — not just a suggestion.

Q1

Stop the Bleeding

- Audit At Risk payment/fraud logs
- Exclude At Risk from retention spend

Target: cancellation rate < 15% or root cause confirmed & written off

Owner: Payments / Fraud

Q1-Q2

Pilot the Nurture Track

- Launch bundles & loyalty incentives
- Target top-frequency Regular customers

Target: 10% of piloted customers shift ≥1 tier by end of Q2

Owner: Lifecycle Marketing

Q2

Protect the Core

- Stand up VIP / white-glove tier
- Proactive outreach on recency drift

Target: High Value segment churn held under 5%

Owner: CRM / VIP Program

Q3

Re-Validate the Model

- Re-run clustering on refreshed data
- Measure Q1-Q2 segment migration

Target: confirm/update segment boundaries; quantify actual lift

Owner: Data / Analytics

Three Scenarios, One Combined Upside

SCENARIO A

Fix At Risk

+\$12K – \$25K

Resolve the root cause of 99.47% cancellations and recover a portion as real revenue.

SCENARIO B

Nurture Regular

+\$55K – \$90K

Shift 10–15% of Regular customers' spend toward High Value behavior.

SCENARIO C

Combined

+\$150K+

Both tracks executed together over 2–3 quarters.

+\$150K+ estimated annual revenue upside — combining Regular → High Value nurture campaigns with eliminating At Risk-segment waste, without new customer acquisition spend.

Three Paths for the Next Four Quarters

Illustrative projection built from segment averages and the roadmap's own targets — not a statistically fitted time-series forecast, but a planning input to compare doing nothing vs. acting.



⚠ Risk of Inaction

- Every quarter of delay on the Regular nurture track ≈ \$13.7K–\$22.5K in foregone revenue
- Unresolved At Risk failure mode could spread if it's systemic (payments/fraud), not isolated
- Without quarterly re-segmentation, targeting accuracy decays as behavior drifts

By Q4, With Roadmap vs. Risk of Inaction:

≈ \$58K / quarter swing

(\$172K vs. \$114K) — the real cost of choosing not to act

Let's Discuss

Which of the four roadmap priorities should we own first this quarter?

A full interactive version of this analysis — live charts, a rotatable 3D cluster view, and every underlying number — is available as a companion report for anyone who wants to dig deeper after this meeting.

Ali Ahmed Zaky · Business Intelligence Expert

KMeans · $k=3$ · Silhouette 0.432 · 909 customers · 13 behavioral features